Project organization

COVER PAGE

File: 01_cover_page.kicad_sch

BLOCK DIAGRAM

File: 02_block_diagram.kicad_sch

CIRCUIT

File: 03_circuit.kicad_sch

MECHANICAL

File: 04_mechanical.kicad_sch

REVISION HISTORY

File: 05_revision_history.kicad_sch

Imported from the Altium Designer project (see altium/)

Sheet: /
File: PALHASERVO.kicad_sch

Title: PALHASERVO

Size: A3	Date:	**Rev: V1l1**
KiCad E.D.A. 10.0.7		Id: 1/6

=Project

=CurrentDate

Assembly variant: Assembly_Variant

Version/Revision

PRELIMINAR

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Project Notes

Note:
General information about the project

Note:
Critical consideration about the project

Note:
Care should be taken

Note:
Critical layout guide

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Sheet: /COVER PAGE/
File: 01_cover_page.kicad_sch

Title: COVER PAGE

Size: A3	Date:	Rev: V1\1
KiCad E.D.A. 10.0.7		Id: 2/6

=project
(Block diagram)

Pages 3

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UART

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HC – 05

PWM

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AN1

CLOCKS (CPU & PCIE)

Imported from the Altium Designer project (see altium/)

Sheet: /BLOCK DIAGRAM/
File: 02_block_diagram.kicad_sch

Title: BLOCK DIAGRAM

Size: A3	Date:	Rev: V1I1
KiCad E.D.A. 10.0.7		Id: 3/6

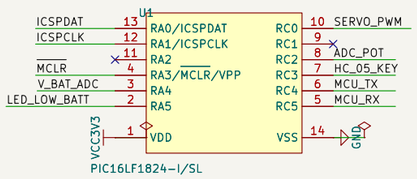
Circuit

MCU

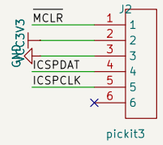
Note:
When selecting pin refer to pin allocation table in the devices datasheet.

Note:
Standard PWM mode is routed to any CCPx pin.
Enhanced PWM mode has five modes:
• Single PWM
• Half-Bridge PWM
• Full-Bridge PWM, Forward Mode
• Full-Bridge PWM, Reverse Mode
• Single PWM with PWM Steering Mode

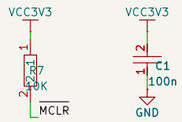
OUTPUTs are routed to PxA, PxB, PxC, PxD pins



ICSP

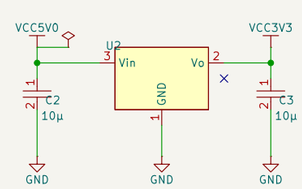


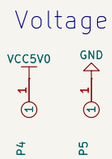
Decoupling



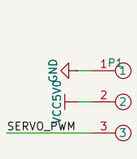
DC-DC REGULATOR

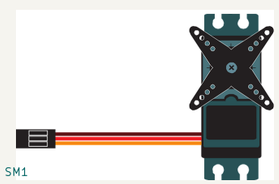
Voltage Input



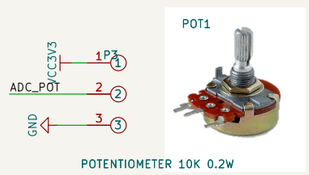


SERVOMOTOR



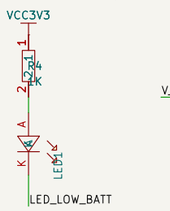


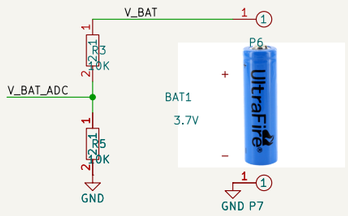
KNOB



BATTERY MANAGEMENT

Low_Battery_LED



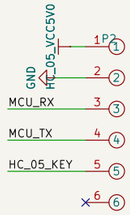


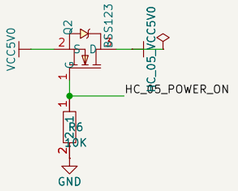
Note:
Li-Ion batteries are sensitive to over-discharge, which is why many cells have build-in under-voltage protection circuits that switch off the cell when the cell is discharged below 2.5V. It is recommended to recharge the battery before this battery internal protection is activated.

source:
http://www.richtek.com/battery-management/en/designing-liion.html

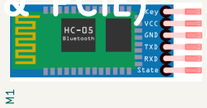
Note:
A resistor divider is used to monitor the battery voltage. MCU is wake from sleep mode periodically and measure battery voltage. If low battery is detected, system will be shut down.

BLUETOOTH SERIAL





Note:
-Hold KEY in high state to enter AT COMMANDS. Baud rate will be 38400.
-AT CMDs are written in uppercase and terminate with CR LF.



CLOCKS (CPU & PCIE)

Imported from the Altium Designer project (see altium/)

Sheet: /CIRCUIT/
File: 03_circuit.kicad_sch

Title: CIRCUIT

Size: A3	Date:	Rev: V1l1
KiCad E.D.A. 10.0.7		Id: 4/6

MECHANICAL

LOGO

Create a PCB logo Component if desired

PCB

PCB1

PCB

Give_The_PCB_a_NAME

PRODUCT CASE

CASE1



POWER BOX CASE

Note:
The power bank case was chosen because of its form factor and have a complete charger solution built-in and finally its price.

Taking advantage of those features acomplishes one of the product's features which is not to have cost with case customization.

Then, some moding is required to use such case:
- Li-ion battery replacement. A smaller Li-ion AA size fits;
- Drill a hole for the knob.

Imported from the Altium Designer project (see altium/)

Sheet: /MECHANICAL/
File: 04_mechanical.kicad_sch

Title: MECHANICAL

Size: A3	Date:	Rev: V1i1
KiCad E.D.A. 10.0.7		Id: 5/6

REVISION HISTORY

VxIx – Write here release notes

CLOCKS (CPU & PCIE)

Imported from the Altium Designer project (see altium/)

Sheet: /REVISION HISTORY/
File: 05_revision_history.kicad_sch

Title: REVISION HISTORY

Size: A3	Date:	**Rev: V1I1**
KiCad E.D.A. 10.0.7		Id: 6/6